Page classification output

How each page's classification is reported. Taxonomy v24 (16 L1, 83 L2, 60 L3, 31 specialties), built from Page_Taxonomy.xlsx.

1. The two reported columns

Every page gets a **Page Type** and a **Page Subtype**, assembled from the classification axes:

Column	Value	Rule
Page Type	L1	When the page has no L3.
	L1 (L2)	When the page has an L3 - only under Lab Report/Result, where an L3 is always present (Others if none of the named ones fits).
Page Subtype	L2	When the page has an L2 and no L3.
	L3	When the page has an L3: the path moves up one level.
	... | Sp: Z	Appended when a specialty Z is known. Specialty **Unknown** (no signal on the page) is left out; **Others** (a service line not on the list) is shown.
	(blank)	L1s with no L2 (Problem List, Others) and no known specialty.
	Sp: Z	L1 with no L2 but a known specialty.

In short: without an L3, `Page Type = L1` and `Page Subtype = L2 [ | Sp: Z ]`; with an L3, `Page Type = L1 (L2)` and `Page Subtype = L3 [ | Sp: Z ]`.

2. Examples

L1	L2	L3	Specialty	Page Type	Page Subtype
Progress Note	CC/HPI	-	Ophthalmology	**Progress Note**	**CC/HPI | Sp: Ophthalmology**
Progress Note	Visit Diagnosis	-	Unknown	**Progress Note**	**Visit Diagnosis**
Lab Report/Result	Diagnostic Study	Echocardiogram Report	Cardiology	**Lab Report/Result (Diagnostic Study)**	**Echocardiogram Report | Sp: Cardiology**
Lab Report/Result	Laboratory	Urinalysis Report	Unknown	**Lab Report/Result (Laboratory)**	**Urinalysis Report**
Lab Report/Result	Radiology	Others	Unknown	**Lab Report/Result (Radiology)**	**Others**
Discharge Summary	Transfer of Care	-	Emergency	**Discharge Summary**	**Transfer of Care | Sp: Emergency**
Demographics	Face Sheet	-	Unknown	**Demographics**	**Face Sheet**
Procedure Note	Therapy Session	-	Others	**Procedure Note**	**Therapy Session | Sp: Others**
Problem List	-	-	Unknown	**Problem List**	**(blank)**
Others	-	-	Unknown	**Others**	**(blank)**

3. What the axes mean

Axis	Decided by	Notes
L1	the page classifier	16 values. A page nothing else claims is Others and goes to an auditor.
L2	the classifier, masked by L1	Only L2s under the page's L1 can occur. Under **Progress Note**, L2 is the content section on the page (CC/HPI, Visit Diagnosis, Signature, ...), from section-header matching, not the kind of note.

Axis	Decided by	Notes
L3	the classifier, masked by L2	Only under Lab Report/Result (Diagnostic Study, Laboratory, Pathology, Radiology). Blank everywhere else, and blank is correct.
Specialty	keyword detector + the document	Rides alongside the path, never part of it. Unknown = no signal; Others = extend the list.
Codeability	canon tag + rules	Reported in its own column, never folded into the subtype.

4. Full output record

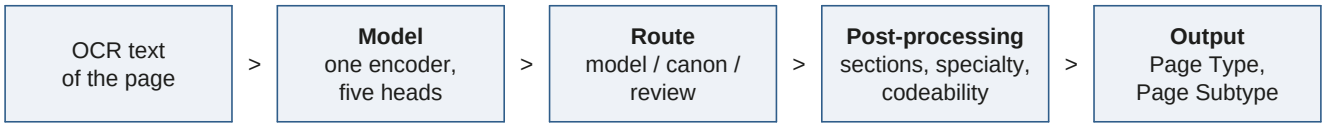
The two display columns sit beside the raw axes, so the display can change without losing information:

Field	Example	Notes
page_type	Lab Report/Result (Diagnostic Study)	Display: rule in section 1
page_subtype	Echocardiogram Report | Sp: Cardiology	Display: rule in section 1
L1 / L2 / L3	Lab Report/Result / Diagnostic Study / Echocardiogram Report	Raw path
specialty	Cardiology	Raw specialty, including Unknown
codeability	codeable	codeable, non_codeable, discharge or needs_review
codeability_rule	page_type_tag	page_type_tag, context (needs continuity), source (needs the ingest source)
confidence / route	0.91 / model	model, canon or review; an unsure page is never forced into a bucket

A page that cannot be decided is reported as Others / needs_review, not as a best guess and never as non_codeable.

5. How the classification is made

Two stages: a trained model reads the page text, then rule-based post-processing turns its predictions into the reported labels and decides when not to trust it.



5.1 The model

- **Input.** The page's OCR text. A long page keeps the first 85% of the 512-token budget from the top and the last 15% from the bottom: the title and letterhead sit at the top, the signature at the foot.
- **One encoder, five outputs.** A DistilBERT encoder feeds five classifiers trained together: L1, L2, L3, specialty and codeability. Each is data-starved alone; sharing one encoder lets them help each other.
- **Hierarchy inside the model.** L2 can only be an L2 of the predicted L1, and L3 only an L3 of the predicted L2; impossible combinations are masked out before the choice is made.
- **What it does not learn.** The L2 under Progress Note (the content section, see 5.2). An L1 is learned once it has more than 10 labelled pages; below that it is grouped as OTHER and left to the canon (today Immunization and Others). L2, L3 and specialty need 3 pages: a rarer L3 becomes Others, a rarer L2 or specialty is left to post-processing. Today: 15 L1, 27 L2, 23 L3, 13 specialties.
- **Where L3 comes from.** The model predicts it, choosing only among the L3s of the L2 it predicted (only Lab Report/Result has L3s). On pages routed to the canon, the canon's page type gives the L3 instead.
- **Training and testing.** 1,011 labelled pages, split by chart so pages of one chart are never on both sides; pages without a chart id stand alone. Reported: accuracy of the full Page Type / Subtype, partial credit when L1 is right, and confusion tables.
- **Output.** For each of the five: a label and a confidence, plus the top three L1s.

5.2 Post-processing, in order

Step	What happens
1. Route	On the L1 confidence. **model**: at least 50% and 10 points ahead of the second choice. **canon**: the model said OTHER, or is at least 25% sure. **review**: anything less. An unsure page is never forced into a bucket.
2. Canon	For pages routed to it: a keyword list per canon page type (298 types, each tied to one path). A title used as a heading near the top counts most, body terms less, generic terms only alongside others. The best page type gives the path.
3. Section	Under Progress Note, L2 is the content section from the page's headers (CC/HPI, Visit Diagnosis, Signature, ...). Visit Diagnosis outranks the rest; a page with no header is Others.
4. L3	Under Lab Report/Result an L3 is required: Others when none of the named ones fits.
5. Specialty	The keyword detector, and the specialty of the document the page belongs to, come first; the model's specialty is the fallback. No signal is Unknown.
6. Codeability	The page type's tag from the canon. Forms, questionnaires and problem lists are codeable inside a progress note and not standing alone, so they wait for the pages to be grouped into documents; billing statements need the ingest source. Until then: needs_review, never non_codeable. The model's codeability is a cross-check.
7. Display	Page Type and Page Subtype as in section 1.

Status (October 2026). Built: the taxonomy, the five-output trainer (run on Colab), the canon and the review tool. To do: wire them into the pipeline's page-classification stage (it still runs the earlier single-output model), and add the specialty detector.

Appendix: every Page Type / Page Subtype

All 141 valid paths, as they print with no specialty. With a specialty Z, each subtype gains ｜ Sp： Z. Under Progress Note, the subtype is the content section.

Page Type	Page Subtype
Progress Note	Assessment and Plan
Progress Note	CC/HPI
Progress Note	History
Progress Note	Imaging Results
Progress Note	Lab Results
Progress Note	Medication List
Progress Note	Patient Instructions
Progress Note	Physical Exam
Progress Note	Plan of Care
Progress Note	Problem List
Progress Note	Procedure
Progress Note	Questionnaire
Progress Note	Signature
Progress Note	Summary
Progress Note	Visit Diagnosis
Progress Note	Others
Lab Report/Result (Diagnostic Study)	Ankle-Brachial Index Report
Lab Report/Result (Diagnostic Study)	Bronchoscopy Report
Lab Report/Result (Diagnostic Study)	Cardiac Catheterization Report
Lab Report/Result (Diagnostic Study)	Carotid Duplex Report
Lab Report/Result (Diagnostic Study)	Colonoscopy Report
Lab Report/Result (Diagnostic Study)	Colposcopy Examination
Lab Report/Result (Diagnostic Study)	Cystoscopy Report
Lab Report/Result (Diagnostic Study)	Echocardiogram Report
Lab Report/Result (Diagnostic Study)	Electrocardiogram
Lab Report/Result (Diagnostic Study)	Electroencephalogram Report
Lab Report/Result (Diagnostic Study)	Electromyography and Nerve Conduction
Lab Report/Result (Diagnostic Study)	Esophageal Manometry Report
Lab Report/Result (Diagnostic Study)	Esophagogastroduodenoscopy Report
Lab Report/Result (Diagnostic Study)	Extremity Doppler Report
Lab Report/Result (Diagnostic Study)	Fundus Photography Report
Lab Report/Result (Diagnostic Study)	Holter Monitor Report
Lab Report/Result (Diagnostic Study)	Laryngoscopy Procedure Report
Lab Report/Result (Diagnostic Study)	Lens Parameters
Lab Report/Result (Diagnostic Study)	Myocardial Perfusion Study
Lab Report/Result (Diagnostic Study)	Neurocognitive Screening Examination
Lab Report/Result (Diagnostic Study)	Oximetry Report
Lab Report/Result (Diagnostic Study)	Pacemaker Device Report
Lab Report/Result (Diagnostic Study)	Pap and HPV Report

Page Type	Page Subtype
Lab Report/Result (Diagnostic Study)	Plethysmography Report
Lab Report/Result (Diagnostic Study)	Pulmonary Function Test
Lab Report/Result (Diagnostic Study)	Sigmoidoscopy Report
Lab Report/Result (Diagnostic Study)	Sleep Study Report
Lab Report/Result (Diagnostic Study)	Split Night Study Report
Lab Report/Result (Diagnostic Study)	Stress Test Report
Lab Report/Result (Diagnostic Study)	Sudoscan Report
Lab Report/Result (Diagnostic Study)	Titration Report
Lab Report/Result (Diagnostic Study)	Videonystagmography Report
Lab Report/Result (Diagnostic Study)	Visual Field Interpretation
Lab Report/Result (Diagnostic Study)	Others
Lab Report/Result (Laboratory)	Allergy Testing Report
Lab Report/Result (Laboratory)	Chemistry Report
Lab Report/Result (Laboratory)	FISH Report
Lab Report/Result (Laboratory)	Genetic Test Report
Lab Report/Result (Laboratory)	Glucose Report
Lab Report/Result (Laboratory)	Hematology Report
Lab Report/Result (Laboratory)	INR and Prothrombin Time
Lab Report/Result (Laboratory)	Laboratory Flow Sheet
Lab Report/Result (Laboratory)	Microbiology Report
Lab Report/Result (Laboratory)	Urinalysis Report
Lab Report/Result (Laboratory)	Others
Lab Report/Result (Pathology)	Biopsy Report
Lab Report/Result (Pathology)	Cytology Report
Lab Report/Result (Pathology)	Cytometry Report
Lab Report/Result (Pathology)	Molecular Diagnostics Report
Lab Report/Result (Pathology)	Others
Lab Report/Result (Radiology)	Bone Mineral Density Report
Lab Report/Result (Radiology)	CT Angiography Report
Lab Report/Result (Radiology)	CT Scan Report
Lab Report/Result (Radiology)	MR Angiography Report
Lab Report/Result (Radiology)	MRI Report
Lab Report/Result (Radiology)	Mammogram Report
Lab Report/Result (Radiology)	Nuclear Medicine Report
Lab Report/Result (Radiology)	Ultrasound Report
Lab Report/Result (Radiology)	X-Ray Report
Lab Report/Result (Radiology)	Others
Operative Note	Anesthesia Record
Operative Note	Critical Care Record
Operative Note	Procedure
Operative Note	Surgery Report
Operative Note	Triage Report

Page Type	Page Subtype
Operative Note	Others
Procedure Note	Dialysis Record
Procedure Note	Elastography
Procedure Note	Infusion and Chemotherapy
Procedure Note	Injection Visit
Procedure Note	Knee Procedure
Procedure Note	Laser Procedure
Procedure Note	Lumbar Procedure
Procedure Note	Office Procedure
Procedure Note	Pathology Procedure
Procedure Note	Radiation Therapy
Procedure Note	Therapy Session
Procedure Note	Wound Care
Procedure Note	Others
Discharge Summary	Depart/Discharge Note
Discharge Summary	Discharge Instruction
Discharge Summary	Discharge Medication
Discharge Summary	Interim Summary
Discharge Summary	Short Stay Summary
Discharge Summary	Transfer of Care
Discharge Summary	Others
Demographics	Face Sheet
Demographics	Patient Demographics
Demographics	Others
Immunization	Immunization Record
Immunization	Others
Medication Record	Blood Transfusion Record
Medication Record	Medication List
Medication Record	Medication Reconciliation
Medication Record	Prescription Page
Medication Record	Refill Record
Medication Record	Others
Order and Requisition	Laboratory Requisition
Order and Requisition	Physician Order
Order and Requisition	Others
Consent Form	HIPAA Form
Consent Form	Immunization Consent Form
Consent Form	Patient Agreement Form
Consent Form	Surgical Consent Form
Consent Form	Transfusion Consent Form
Consent Form	Others
Authorization	Assignment of Benefits

Page Type	Page Subtype
Authorization	Prior Authorization Request
Authorization	Referral Authorization
Authorization	Release of Information Authorization
Authorization	Others
Problem List	(blank)
Forms and Questionnaires	Assessment
Forms and Questionnaires	Questionnaire
Forms and Questionnaires	Others
Patient Communication	Notice
Patient Communication	Patient Education Material
Patient Communication	Patient Message
Patient Communication	Provider Letter
Patient Communication	Others
Administrative	Administrative Record
Administrative	Appointment Record
Administrative	Billing Statement
Administrative	Others
Others	(blank)